GEM-PIC-RZ: a Galilean electromagnetic particle-in-cell code in quasi-cylindrical geometry with free-hand grids

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Plasma accelerators couple the micrometre wavelength of the driving laser to acceleration lengths of millimetres to metres. Full particle-in-cell (PIC) codes resolve the laser but are bound by the Courant condition; quasi-static codes are fast but lose the laser carrier. We present GEM-PIC-RZ, a quasi-cylindrical implementation of the Galilean electromagnetic PIC scheme. It keeps the slow evolution of the forward field, so it resolves the laser carrier, yet its propagation step is set by the evolution of the laser and the plasma, not by the cell size. Free-hand non-uniform grids and a charge-conserving regularization of the bubble closure make it accurate and robust. Against the full PIC code FBPIC it reproduces the laser wakefield, the depletion and red-shift of a few-cycle pulse, where an envelope model fails, and ionization injection, at about 2 % of the CPU time; the injected charge carries a model uncertainty of about 15 % from the regularization. Self-injection in a sharp density down-ramp is not yet reproduced and is identified as an open problem.

I. INTRODUCTION

Plasma-based accelerators driven by lasers [1, 2] or particle beams sustain fields a thousand times stronger than radio-frequency cavities, and their design rests on simulations. These simulations are multiscale: the laser wavelength $\lambda_0 \sim 1\,\mu$m, the plasma wavelength $\lambda_p \sim 10$–$100\,\mu$m and the acceleration length of millimetres to metres span six orders of magnitude. Full electromagnetic PIC codes resolve λ_0 and are bound by the Courant condition, so the number of time steps is the acceleration length divided by a fraction of λ_0. Quasi-cylindrical (azimuthal-mode) codes such as Calder-Circ and FBPIC [3, 4] reduce the cost per step, and the Lorentz-boosted frame [5] and Galilean spectral solvers [6] reduce the number of steps or suppress the numerical Cherenkov instability, but the Courant limit remains.

Quasi-static codes [7–14] remove it by dropping *all* derivatives with respect to the slow time. The plasma response of a slice then follows from the sources of that slice, and the driver is advanced with a step set by its own evolution. The price is the laser carrier: lasers enter through ponderomotive envelope models [8, 15], and ionization in the instantaneous field, few-cycle pulses with strong red-shift, harmonics and the radiation of the accelerated beam are outside the model.

The Galilean electromagnetic particle-in-cell (GEM-PIC) algorithm [16] closes this gap. It transforms Maxwell's equations and the Vlasov equation to the Galilean coordinates $\zeta = z - ct$ and a slow coordinate s, either $s = ct$ (GT1) or $s = z$ (GT2), and splits the transverse field into forward and backward components. The slow derivative is kept for the forward component, which carries the laser, so the carrier is resolved and propagates, diffracts, depletes and red-shifts self-consistently. It is dropped only for the backward and longitudinal components, where it appears together with the much larger ζ-derivative. The scheme is therefore *not* a quasi-static approximation, yet it has no Courant limit, because the field update is semi-implicit. In three dimensions GEM-PIC reproduced a 31.8 cm laser wakefield accelerator with ionization injection simulated with FBPIC [16].

Most laser and beam drivers are close to axisymmetric, and a few azimuthal modes describe them at a small fraction of the cost of a three-dimensional run. This paper presents GEM-PIC-RZ, the implementation of GEM-PIC in this geometry, and asks how far it can replace a full PIC code in the problems that matter for plasma accelerators: the nonlinear wake, the evolution of the laser over many Rayleigh lengths, and the injection of electrons. Section II gives the reduced equations in azimuthal modes and the particle scheme. Section III describes the discretization on non-uniform radial grids, its dispersion relation, and two numerical problems with their remedies: a numerical Cherenkov radiation behind relativistic bunches and the excitation of spurious modes by the singular bubble closure. Section IV summarizes the implementation. Section V verifies the code against analytic references, and Sec. VI compares it with FBPIC for four laser-plasma problems, three of which it reproduces and one, density down-ramp injection, which it does not yet. Section VII outlines the extension to free-electron lasers.

Like QUARZ [35], GEM-PIC-RZ is a *zero-touch* code: every line of its source code, of its test suite, of its benchmark scripts and of its documentation was written by an AI agent, the large language model Claude [39], working with a compiler, a shell and the test suite. The author directed the work without writing or editing a single line of code. The human contributions were those that require physical judgment: the GEM-PIC formulation itself [16], the choice of methods and of the problems to solve, physical conventions (for example, the Galilean transformation used for the particles), the criteria that decide whether a result can be trusted, and the interpretation of the results. The agent worked under a fixed rule: no feature counts as done before it passes a test against an independent analytic or numerical reference, and every change is

rerun against the whole suite, so the correctness of the code rests on tests rather than on line-by-line human review. Several numerical errors described in this paper (the numerical Cherenkov radiation, the closure at the rear of the window, the self-fields of bunches with $v < c$, the undulator kick and the timing of the beam current) were found in this way, by tests that failed against their references. Every commit in the public repository [40] records the AI session that produced it, so the claim can be checked.

II. MODEL

This section states the equations that GEM-PIC-RZ solves: the reduced Maxwell equations of GEM-PIC in azimuthal modes and the motion of the particles in the Galilean variables. The derivation in Cartesian geometry is given in Ref. [16]; here we show how it carries over to cylindrical geometry and which choices the code makes.

A. Variables and field equations

We use units with $c = 1$, lengths in $1/k_0$, time in $1/\omega_0$ ($k_0 = \omega_0/c$ the laser wavenumber), densities in the critical density n_c, momenta in $m_e c$ and fields in $m_e c\,\omega_0/e$. Following Ref. [16], the fast coordinate is $\zeta = z - t$, and the slow coordinate is either $s = t$ (GT1, $\partial_t = \partial_s - \partial_\zeta$, $\partial_z = \partial_\zeta$) or $s = z$ (GT2, $\partial_t = -\partial_\zeta$, $\partial_z = \partial_s + \partial_\zeta$). The reduced field equations derived below are the same for both; the particles of GEM-PIC-RZ are advanced in GT2 (Sec. II B), so a level s of the simulation is the plane $z = s$ of the laboratory, and the slice ζ of that level is the time $t = s - \zeta$ at which the fields pass the plane. The simulation window covers $0 \le \zeta \le L_w$; its front is at $\zeta = L_w$. Every field is expanded in azimuthal modes,

$$F(r,\varphi,\zeta,s) = F_0 + 2\,\mathrm{Re}\sum_{m=1}^{M} F_m(r,\zeta,s)\,e^{im\varphi}, \qquad (1)$$

and we drop the mode index below. A linearly or circularly polarized laser lives in $m = 1$; $m = 0$ carries the wake.

The transverse fields are combined into the characteristic components

$$R_r = E_r + B_\varphi, \qquad L_r = E_r - B_\varphi,$$
$$R_\varphi = E_\varphi - B_r, \qquad L_\varphi = E_\varphi + B_r. \qquad (2)$$

A wave travelling in $+z$ at the speed of light has $L = 0$ and is a function of ζ only; R is the forward (laser) field. With GT1, Maxwell's equations become, exactly,

$$\partial_s R_r = \partial_r E_z + \frac{im}{r} B_z - j_r, \qquad (3)$$

$$\partial_s R_\varphi = \frac{im}{r} E_z - \partial_r B_z - j_\varphi, \qquad (4)$$

$$(\partial_s - 2\partial_\zeta) L_r = -\Big(\partial_r E_z - \frac{im}{r} B_z + j_r\Big), \qquad (5)$$

$$(\partial_s - 2\partial_\zeta) L_\varphi = -\Big(\partial_r B_z + \frac{im}{r} E_z + j_\varphi\Big), \qquad (6)$$

$$(\partial_s - \partial_\zeta) B_z = -\frac{1}{r}\partial_r (r E_\varphi) + \frac{im}{r} E_r, \qquad (7)$$

$$(\partial_s - \partial_\zeta) E_z = \frac{1}{r}\partial_r (r B_\varphi) - \frac{im}{r} B_r - j_z. \qquad (8)$$

The forward field varies fast in ζ (on λ_0) and slowly in s (on the Rayleigh length and the depletion length). The backward field L and the longitudinal fields E_z, B_z of a co-propagating structure vary on the same ζ scales as their sources, which move with $v \approx c$, so their s-derivatives are small compared with their ζ-derivatives. With GT2 the same equations hold with $\partial_s - 2\partial_\zeta \to -\partial_s - 2\partial_\zeta$ in Eqs. (5), (6) and without ∂_s in Eqs. (7), (8) [16]. Neglecting ∂_s wherever it appears together with ∂_ζ gives, for both transformations, the reduced system

$$\partial_\zeta L_r = \tfrac{1}{2}\Big(\partial_r E_z - \frac{im}{r} B_z + j_r\Big), \qquad (9)$$

$$\partial_\zeta L_\varphi = \tfrac{1}{2}\Big(\partial_r B_z + \frac{im}{r} E_z + j_\varphi\Big), \qquad (10)$$

$$\partial_\zeta B_z = \frac{1}{r}\partial_r (r E_\varphi) - \frac{im}{r} E_r, \qquad (11)$$

$$\partial_\zeta E_z = -\frac{1}{r}\partial_r (r B_\varphi) + \frac{im}{r} B_r + j_z, \qquad (12)$$

together with Eqs. (3), (4), where $E = (R + L)/2$, $B_\varphi = (R_r - L_r)/2$ and $B_r = (L_\varphi - R_\varphi)/2$. These are the azimuthal-mode form of the universal system, Eqs. (49)–(54) of Ref. [16]: with $(x, y) \to (r, \varphi)$, $ik_x \to \partial_r$ (or $r^{-1}\partial_r r$ in the curls) and $ik_y \to im/r$, the equations agree term by term. Equations (9)–(12) are integrated from the front of the window, where $L = E_z = B_z = 0$, towards the rear. In vacuum the system gives $\omega = 2\kappa k_\perp^2/(4\kappa^2 - k_\perp^2) \simeq k_\perp^2/(2\kappa)$ for a mode $\propto e^{i\kappa\zeta - i\omega s}$, the paraxial limit of the exact $\omega = \sqrt{\kappa^2 + k_\perp^2} - \kappa$; in a cold plasma $k_\perp^2$ is replaced by $k_\perp^2 + \omega_p^2$ [16]. The backward wave is not described; Ref. [16] shows that the forward branch of the full dispersion relation approaches that of the reduced system for $\kappa \gg \omega_p$.

B. Particles

The particles follow the characteristics of the Vlasov equation in GT2 [16],

$$\frac{\mathrm{d}s}{\mathrm{d}\zeta} = \frac{v_z}{v_z - 1}, \qquad \frac{\mathrm{d}\boldsymbol{r}_\perp}{\mathrm{d}\zeta} = \frac{\boldsymbol{v}_\perp}{v_z - 1}, \qquad \frac{\mathrm{d}\boldsymbol{p}}{\mathrm{d}\zeta} = \frac{\boldsymbol{F}}{v_z - 1}, \qquad (13)$$

with the Lorentz force $\boldsymbol{F}$ of the fields reconstructed from the modes. A plasma particle at rest keeps its $s = z$ and crosses the whole window within one level, as in the lab frame. A particle whose s reaches the next level s^1 before it leaves the window stays inside it: it is trapped. In principle all particles can be integrated along ζ; as recommended in Ref. [16], GEM-PIC-RZ integrates the fast, trapped ones along s instead, which is explicit and more accurate. Each particle is thus in one of two lists.

Streaming particles: the plasma electrons, mobile ions and electrons freed by ionization. A fresh layer of plasma is loaded at the front of the window at every step $s^0 \to s^1 = s^0 + \Delta$, with the density $n(s^1)$ and a quiet start in r^2 and φ; one macroparticle represents the plasma of a ring of axial thickness Δ. The layer then streams through the whole window, using ζ as the independent variable, Eq. (13). Between two slices with the spacing h_n a particle is pushed over the time

$$\mathrm{d}t = \frac{h_n}{1 - v_z}, \tag{14}$$

by a two-pass Boris push [25] that makes $\mathrm{d}t$ consistent with the mean v_z over the interval; its lab position $z = s$ advances by $v_z\,\mathrm{d}t$. It deposits its current with the weight $W = q\,\mathrm{d}t/\Delta$, the charge that crosses the cell during one step [Eq. (99) of Ref. [16]]. The fields are those of the level s^1 being computed (Sec. III B).

Beam particles: bunches and trapped particles, which move with the window. They are advanced in s from s^0 to s^1 ($\mathrm{d}t = \mathrm{d}s/v_z$, $\mathrm{d}\zeta = (v_z - 1)\mathrm{d}t$) with the Vay pusher [26] in sub-steps of at most $\Delta z_{\max}$, in the fields of level s^0; the Vay pusher keeps the cancellation $\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B} \approx 0$ of a relativistic beam in its own field at any step, which the Boris pusher does not. A beam particle deposits a current with the weight q/v_z, linear in ζ.

The two lists exchange particles. A streaming particle whose lab position reaches $z = s^1$ inside the window has been trapped; it joins the beam list if $v_z \geq 0.9$, where the push along ζ becomes stiff [$\mathrm{d}\zeta/\mathrm{d}t = v_z - 1 \to 0$]. A slower one, which may happen in a strong laser field where electrons move forward by more than Δ while crossing the pulse, continues to stream in the next step from its ζ. Conversely, a beam particle that becomes slower than $v_z = 0.8$ returns to streaming. No particle is deleted, so the charge is conserved.

C. Initial fields, ionization and ions

A laser pulse is initialized in mode $m = 1$ as a forward wave, $R_{r,1} = P^*$, $R_{\varphi,1} = iP^*$, $L = 0$, with $P = E_x + iE_y$ a paraxial Gaussian beam (with wavefront curvature and Gouy phase at a distance from focus) or a given radial profile.

A bunch of velocity β starts with its own co-moving field, $-(\nabla_\perp^2 + \gamma^{-2}\partial_\zeta^2)\psi = j_z/\beta$, $R = (1 + \beta)\boldsymbol{E}_\perp$, $L = (1 - \beta)\boldsymbol{E}_\perp$, $E_z = -\gamma^{-2}\partial_\zeta\psi$, with $\boldsymbol{E}_\perp = -\nabla_\perp\psi$. The problem is solved with the radial eigenvectors of the discrete Laplacian of the field solver and one tridiagonal solve along ζ per eigenvector, so a bunch in vacuum is an exact discrete equilibrium. A field that is not the co-moving one would stay behind a bunch with $v < c$ that slips back in the window and exchange energy with it; this matters for the FEL application (Sec. VII).

Ion species can be mobile, with their own density profile (e.g. a dopant section), and field-ionizable. The ADK tunnel rate [27, 28] of each level is evaluated with the instantaneous $|\boldsymbol{E}|$, including the laser carrier, and averaged over the two slices of the interval; ionization energies are taken from the NIST database [29]. A Monte Carlo step with exponential waiting times allows several levels per interval. The random numbers come from a counter-based hash of (seed, particle, step, slice, pass), so all back-ends draw the same numbers.

III. DISCRETIZATION

The discretization has to meet three requirements. It must keep the constraints of Maxwell's equations on a radial grid that is fine at the axis and coarse outside, since the plasma response and the laser are concentrated near the axis while the fields extend far out. It must keep each slice cheap, since a slice is solved once or twice per step for every one of thousands of slices. And it must be stable for steps Δ much larger than the cell size. This section shows how the scheme meets them and which numerical modes it leaves, and how they are controlled.

A. Radial grid and operators

The radial grid has faces ρ_i, $i = 0, \ldots, N$ ($\rho_0 = 0$ is the axis), and centres c_i, $i = 0, \ldots, N-1$, $\rho_i < c_i < \rho_{i+1}$. Faces carry R_r, L_r, B_z, j_r; centres carry R_φ, L_φ, E_z, j_φ, j_z. With the finite-volume measures

$$A_i^f = \rho_i(c_i - c_{i-1}), \quad A_0^f = \tfrac{1}{2}c_0^2, \qquad A_i^c = c_i(\rho_{i+1} - \rho_i), \tag{15}$$

the operators are

$$(Ge)_i = \frac{e_i - e_{i-1}}{c_i - c_{i-1}}, \qquad (gb)_i = \frac{b_{i+1} - b_i}{\rho_{i+1} - \rho_i}, \tag{16}$$

$$(\mathrm{CE})_i = \frac{c_i E_{\varphi,i} - c_{i-1}E_{\varphi,i-1}}{A_i^f} - \frac{im}{\rho_i}E_{r,i}, \tag{17}$$

$$(\mathrm{CB})_i = \frac{\rho_{i+1}B_{\varphi,i+1} - \rho_i B_{\varphi,i}}{A_i^c} - \frac{im}{c_i}B_{r,i}, \tag{18}$$

for ∂_r to the faces and to the centres and for the two curls in Eqs. (11), (12). Two discrete identities hold exactly on any grid,

$$\mathrm{CE}\big(Ge,\ \tfrac{im}{c}e\big) = 0, \qquad \mathrm{CB}\big(\tfrac{im}{\rho}b,\ -gb\big) = 0, \tag{19}$$

because $A_i^f = \rho_i(c_i - c_{i-1})$ and $A_i^c = c_i(\rho_{i+1} - \rho_i)$; they are the discrete forms of $\nabla \times \nabla = 0$ in the two

staggerings. The divergence on the measures A^c makes the discrete Gauss law exact. The particles deposit with area weighting plus a quadratic correction per interval, $w = x + \kappa x(1-x)$ with x linear in r^2, where κ is chosen so that the deposited volumes equal A^f, A^c: a uniform plasma is deposited exactly, also on a stretched grid. Fields are gathered linearly in r, which is exact near the axis for fields $\propto r$.

B. The march in ζ

The window has slices $\zeta_0 = 0 < \zeta_1 < \cdots < \zeta_{N_\zeta}$ with spacings $h_n = \zeta_{n+1} - \zeta_n$. Fields live on the slices, currents on the intervals (half-cells) between them. A step computes the level s^1 from s^0 by marching from the front slice to the rear. At slice n the slice $n+1$ of level s^1 is known.

Equations (3), (4) are integrated with the trapezoidal rule in s,

$$R^1_n = R^0_n + \frac{\Delta}{2}\left[\mathcal{F}(E_z, B_z)^0_n + \mathcal{F}(E_z, B_z)^1_n\right] - \frac{\Delta}{4}\,\mathcal{J}_n, \quad (20)$$

where $\mathcal{F}$ is the right-hand side without the current and $\mathcal{J}_n$ is the sum of the currents of the two adjacent half-cells at both levels. On a non-uniform ζ grid the current at the slice is the linear interpolation $(h_n j_{n-1/2} + h_{n-1} j_{n+1/2})/(h_{n-1} + h_n)$. Equations (9)–(12) are integrated with the trapezoidal rule over the half-cell h_n, e.g.

$$L^1_n = L^1_{n+1} - \frac{h_n}{4}\left[\mathcal{G}^1_{n+1} + \mathcal{G}^1_n\right] - \frac{h_n}{2}\, j_{n+1/2}. \quad (21)$$

R^1_n and L^1_n are linear in the unknown E_z and B_z of slice n. Inserted into the trapezoidal rules for Eqs. (11), (12), the identities (19) remove E_z from the B_z equation and B_z from the E_z equation. What remains are two independent tridiagonal systems per mode,

$$\left[16 - h_n(2\Delta + h_n)\,\nabla^2_{f,m}\right] B_{z,n} = \text{known}, \quad (22)$$

$$\left[16 - h_n(2\Delta + h_n)\,\nabla^2_{c,m}\right] E_{z,n} = \text{known} - 16 h_n\, j_{z,n+1/2}, \quad (23)$$

with the discrete Laplacians on the faces and on the centres. R and L then follow explicitly. The matrices depend only on m, Δ and h_n; they are factorized once per distinct h_n and recomputed only when Δ changes. The cost of a slice is linear in N and M.

The currents of half-cell $n+\frac{1}{2}$ at level s^1 depend on the fields of slice n, which are being computed. The march therefore uses a predictor–corrector: the predictor takes the fields of slice n from the old level, $F^1_n \approx F^0_n$ (the slow evolution in s), the streaming particles are pushed through the half-cell and deposit, the slice is solved, and the push and solve are repeated with the new fields (two passes by default). When the half-cell is final, R at slice $n+1$ is corrected for the difference between the predicted and the final current that entered its Eq. (20). At the rear slice the half-cell behind the window is set equal to the last one (zero gradient); a zero current there would halve the source of R and produce a growing spike. Near the outer radial boundary an optional absorber damps R.

C. Free-hand grids

The physics of a plasma accelerator is concentrated in small regions: near the axis, at the laser pulse, at a witness bunch, in a density ramp. All three axes therefore use the same mapping: the step $h(x)$ is given at reference points and is linear in between, and the nodes follow from $\mathrm{d}x/\mathrm{d}k = h(x)$ with k the cell or step index. Within a segment the steps grow geometrically and the nodes are analytic. All slice equations above hold for any h_n, and a uniform grid is the special case of a constant table and is reproduced bit for bit.

D. Numerical dispersion

The step Δ is meant to be orders of magnitude larger than the cell size, so the stability of the scheme has to be shown, not assumed. For a single vacuum Fourier mode $\propto e^{ik_\zeta \zeta - i\omega s}$ with transverse wavenumber $k_\perp$ ($\nabla^2_\perp \to -k^2_\perp$) on a uniform ζ grid, the scheme of Sec. III B gives

$$\tan\frac{\omega\Delta}{2} = \frac{\Delta\, h\, k^2_\perp \sin\theta}{16\sin^2(\theta/2) - h^2 k^2_\perp \cos^2(\theta/2)}, \qquad \theta = k_\zeta h. \quad (24)$$

This is the relation of the 1D plasma model of Ref. [16], Eq. (84) there, $\tan(\omega\Delta/2) = \omega_p^2 \Delta h \sin\theta/[16\sin^2(\theta/2) - \omega_p^2 h^2]$, with ω_p^2 replaced by $k^2_\perp$; the factor $\cos^2(\theta/2)$ comes from the average of the transverse fields over the two slices of a half-cell in the RZ scheme. ω is real for all k_ζ, $k_\perp$, Δ and h, so the scheme has no growing modes in vacuum, and there is no Courant condition linking Δ to h or to the radial cells. With a plasma the phase $\omega\Delta/2$ passes $\pi/2$ where the denominator changes sign, which gives the practical limit $\Delta h \le 4/\omega_p^2$ of Ref. [16]; in typical laser wakefield runs ($\omega_p^2 = 10^{-3}$, $h = 0.25$) it allows Δ of order 10^4. For $\theta \to 0$ and $\Delta \to 0$, $\omega \to 2k_\zeta k^2_\perp/(4k^2_\zeta - k^2_\perp) \simeq k^2_\perp/(2k_\zeta)$, the continuum relation of the reduced system and the paraxial diffraction of the exact $\omega = \sqrt{k^2_\zeta + k^2_\perp} - k_\zeta$. The step Δ is therefore limited by accuracy: by the Rayleigh length of the laser, the plasma and betatron evolution, and in an undulator by the period.

E. Numerical Cherenkov radiation

Equation (24) also shows a defect: at the ζ-Nyquist wavenumber, $\theta = \pi$, $\omega = 0$ for every $k_\perp$. This odd–even mode is stationary in the window, like a relativistic

bunch, so the macroparticle noise of the bunch current drives it resonantly; its group velocity $\mathrm{d}\zeta/\mathrm{d}s = -h^2 k_\perp^2/8$ carries it slowly backwards. The result is a numerical Cherenkov radiation [30] behind the bunch, with $E_x = B_y$ (pure R) and a sign change from slice to slice. It grows with the propagation distance and the shot noise, and a smaller Δ does not help. A $(1,2,1)/4$ filter along ζ applied to the current of the beam particles, whose response $\cos^2(\theta/2)$ vanishes at $\theta = \pi$, removes the source. The filter acts in index space, so it removes the mode at any local spacing of a free-hand grid. The plasma current is not filtered, so the wake is unchanged (Sec. V B).

F. Regularization of the bubble closure

The continuum relation of the reduced system in a cold plasma, $\omega = 2\kappa\omega_p^2/(4\kappa^2 - \omega_p^2)$ [16], has a resonance at $\kappa = \omega_p/2$, where $\omega \to \infty$; it is the remnant of the backward branch, which the reduced system does not describe. The trapezoidal rule in s maps such modes onto modes with $\omega\Delta \to \pi$, which change sign from step to step and are not damped. In a uniform plasma they lie at wavelengths longer than the wake period and are not excited. Where the local density is very high, however, ω_p moves the resonance to resolved scales, and the small, step-dependent jumps of the current that every predictor–corrector and every switch of a particle between levels produce can excite it. In a cold plasma this happens at the back of the bubble (Sec. VI B).

As in QUARZ [35], we do not damp the modes but remove the singularity that excites them. After the deposit of every half-cell, the currents and densities of the streaming plasma (electrons, ions and freed electrons) are smoothed radially, per mode,

$$\begin{aligned} \boldsymbol{j}_\perp &\leftarrow (1 - a^2\nabla_{\mathrm{vec}}^2)^{-1}\boldsymbol{j}_\perp, \\ (\rho, j_z) &\leftarrow (1 - a^2\nabla_m^2)^{-1}(\rho, j_z), \end{aligned} \tag{25}$$

which gives every ring of charge a finite width a. In contrast to QUARZ, which filters every Cartesian component with the scalar operator, $\nabla_{\mathrm{vec}}^2 = \mathrm{grad}\,\mathrm{div} - \mathrm{rot}\,\mathrm{curl}$ is built from the operators of Sec. III: grad is $(G, im/c)$, the curl is CE, $\mathrm{rot}\, b = (imb/\rho, -gb)$, and $\nabla_m^2 = \mathrm{div}\,\mathrm{grad}$ is the Laplacian of the Poisson problem. Because $\mathrm{div}\,\mathrm{rot} = 0$ holds exactly on the staggered grid (with curl $= 0$ on the axis for $m \geq 1$), $\mathrm{div}\,\nabla_{\mathrm{vec}}^2 = \nabla_m^2\,\mathrm{div}$. The filter therefore commutes with the discrete divergence, and the smoothed currents conserve the smoothed charge. The vector system is pentadiagonal (j_r and j_φ interleaved) and the scalar one tridiagonal; both are factorized once per mode, and the filter costs less than 1 % of the run time. The beam particles and the fields acting on the particles are not smoothed. The length a can be chosen separately for the wake ($m = 0$) and for the laser-driven currents ($m \geq 1$).

The smoothing length is a numerical parameter. It stands for the finite width that the thermal spread and deviations from axial symmetry give the returning electron sheath in a real plasma, but it is not derived from them, so every result that may depend on it has to be checked. Two conditions bound it. From below, the fields behind the closure must converge with the step Δ. From above, a must be small compared with the scales of the physics, since it changes the model by $O(a/\ell)$: for the wake ($m = 0$) ℓ is the bubble radius $R_b \approx 2\sqrt{a_0}\, c/\omega_p$, for the laser-driven currents ($m \geq 1$) the laser wavelength and waist. We take the smallest a that meets the first condition and vary it by a factor of two in both directions; Sec. VI F summarizes the result.

IV. IMPLEMENTATION

A scheme becomes a tool only when it runs on the machines where production runs are done and when its results can be trusted and reused. GEM-PIC-RZ is written in C++17 without external dependencies apart from optional MPI and HDF5 libraries. The input is a structured deck in physical units, with profiles of the plasma density, the laser, the bunch current and an undulator taper given as expressions; unknown keys, wrong units and inconsistent settings are reported with file and line. Bunches can also be read from openPMD files of other codes. External elements (quadrupoles, solenoids, active plasma lenses, chicanes, planar and helical undulators) act on the particles only.

Parallelism. A parallel engine runs the serial step as kernels written once behind a thin portability layer; it compiles for OpenMP threads and for NVIDIA GPUs and reproduces the serial step bit for bit (OpenMP) or to 10^{-9} (emulated GPU paths). [GPU timings: the CUDA build compiles and its paths are tested in emulation; measurements on a GPU are pending.] Within one step the march is sequential in ζ, so the window is split into blocks of slices, one per MPI rank, and the ranks form a pipeline in the steps [32]: while rank q computes step k, rank $q+1$, ahead of it in ζ, already computes step $k+1$. The efficiency is $N_s/(N_s + P - 1)$ for N_s steps and P ranks; MPI results agree with the serial run to rounding.

Output and tests. Checkpoints hold the complete state, and a restart is exact on any number of ranks and back-ends. openPMD files [33] hold the fields interpolated to a uniform grid and the exact data on the native non-uniform nodes, in the form we have proposed, together with QUARZ [35], for the openPMD 2.0 standard [34]. A test suite of 25 tests, with about 230 unit checks, runs in minutes and includes all tests of Sec. V.

V. VERIFICATION

Before comparing with other codes, the building blocks have to be shown to be correct against references that do not depend on any simulation code. Table I lists these tests: the self-field of a relativistic bunch against Gauss's

law on uniform and stretched grids, laser diffraction in vacuum, the collapse of ions in a dense bunch, and field ionization against rate equations.

TABLE I. Verification tests (all in the test suite).

test	result
Field of an ultra-relativistic bunch in vacuum vs Gauss's law (uniform, stretched and free-hand radial grids)	E_r to 5×10^{-4}; noise $E_z < 2 \times 10^{-3} E_r$
Same on a free-hand ζ grid (2.5–8 k_0^{-1} across the bunch)	2×10^{-3} (shot-noise level)
$\gamma = 100$ bunch in its own field, 8 cm	energy kept to 2×10^{-5}
Laser focusing in vacuum ($w_0 = 5\,\mu$m, $z_R = 98\,\mu$m, 0.4 mm)	$a_0(z)$ within 3 %
Ion collapse in a dense bunch ($n_b = 100 n_0$, H) [36] vs test-ion model	on-axis n_i 1–3 %; D/H delay 1.418 ($\sqrt{2}$)
Field ionization of H vs $1 - \exp(-\int w\,\mathrm{d}t)$	0.02294 vs 0.02288
$N^{5+} \to N^{6+} \to N^{7+}$ ($a_0 = 2.2$) vs rate equations	0.9357/0.0600/0.0044 vs 0.9367/0.0593/0.0041
ADK rate vs Landau rate (H-like ions)	10^{-12}
Immobile limit of ions ($M = 10^{15}$ u)	identical fields

A. Free-hand ζ and s grids

The free-hand grids are meant to save computing time without loss of accuracy. The test is a beam-driven wake with two short bunches far apart, the configuration of a plasma wakefield accelerator with a driver and a witness: a plasma of $0.002\,n_c$, a driver of $\sigma_z = 1.5\,k_0^{-1}$ and a witness $130\,k_0^{-1}$ behind (Table II). A uniform coarse grid ($h = 2$) is 16 % off at the driver; a free-hand grid that is fine ($h = 0.25$) only at the two bunches reproduces the fine uniform grid to 10^{-3} with a quarter of the slices and 3.6 times faster. In the laser wakefield case of Sec. VI A, a step in s that is small only in the density up-ramp gives the witness energy within 0.4 % of a run with uniformly small steps, with 44 instead of 80 steps; uniform large steps are 1.4 % off.

B. Numerical Cherenkov radiation

Relativistic bunches that propagate over long distances, in a plasma wakefield accelerator or an undulator, are where numerical Cherenkov radiation would spoil a run. A bunch of 1 pC at $\gamma = 1000$ ($\sigma_r = 2$, $\sigma_z = 4\,k_0^{-1}$, 16 slices) propagates 1000 steps of $\Delta = 20\,k_0^{-1}$ in vacuum. Without the filter of Sec. III E the ζ-Nyquist part of the field energy grows to 8.6×10^{-2} of the total; with the filter it stays at 1.3×10^{-4}, the floor of the estimator. In a plasma wake ($0.002\,n_c$, 0.2 pC driver) the Nyquist fraction grows from 1.3 to 1.9×10^{-2} between 200 and 400 steps without the filter and stays at 8×10^{-5} with it, while the bunch parameters agree to four digits and the wake to 0.35 %.

TABLE II. Beam-driven wake on uniform and free-hand ζ grids. E_z errors relative to $\max|E_z|$, witness energy change over 50 steps.

ζ grid	slices	E_z error[a]	ΔE error
uniform $h = 0.25$ (ref.)	2800	–	–
uniform $h = 2$	350	16 % / 6.6 %	1.3 %
free-hand 0.25 / 2	734	0.11 % / 0.84 %	0.06 %

[a] At the driver / at the witness.

VI. LASER-PLASMA ACCELERATION: COMPARISON WITH FULL PIC

The purpose of GEM-PIC-RZ is to replace full PIC simulations of laser-plasma accelerators where the physics allows it. We therefore compare with the quasi-cylindrical spectral full PIC code FBPIC [4] for the four elements of a laser wakefield accelerator: the nonlinear wake in the bubble regime, including the region where the bubble closes, the evolution of the laser over many Rayleigh lengths, and the injection of electrons, by ionization and by a density down-ramp. In all cases FBPIC uses two azimuthal modes and a cell size of $\lambda_0/24$ to $\lambda_0/32$ along z. For the comparisons it matters that a level s of GEM-PIC-RZ is the plane $z = s$ and its slices are the times at which the laser and the wake pass that plane (GT2), while an FBPIC output is a snapshot at one time. Wakes are compared on the axis as functions of ζ, and trapped electrons at the same lab position of the bunch. All decks and analysis scripts are in the repository.

A. The wake in the bubble regime

Most laser wakefield accelerators operate in the bubble (blowout) regime [17, 18], where the ponderomotive force of the laser expels the plasma electrons and leaves a cavity of ions behind the pulse. Its fields accelerate and focus the injected electrons, and its shape and length decide the energy gain. The configuration is a typical one for a 100-TW-class laser: $\lambda_0 = 0.8\,\mu$m, $a_0 = 2$, $w_0 = 8.02\,\mu$m and a field duration of 10.6 fs, linearly polarized and starting at focus, in a plasma of $3.48 \times 10^{18}\,\mathrm{cm}^{-3}$ ($0.002\,n_c$) after a $20\,\mu$m linear ramp, with immobile ions. GEM-PIC-RZ uses 100 radial cells, $0.3\,k_0^{-1}$ on the axis, stretched to $3\,k_0^{-1}$ outside, $h = 0.25\,k_0^{-1}$, $\Delta = 2.5\,\mu$m (a hundred

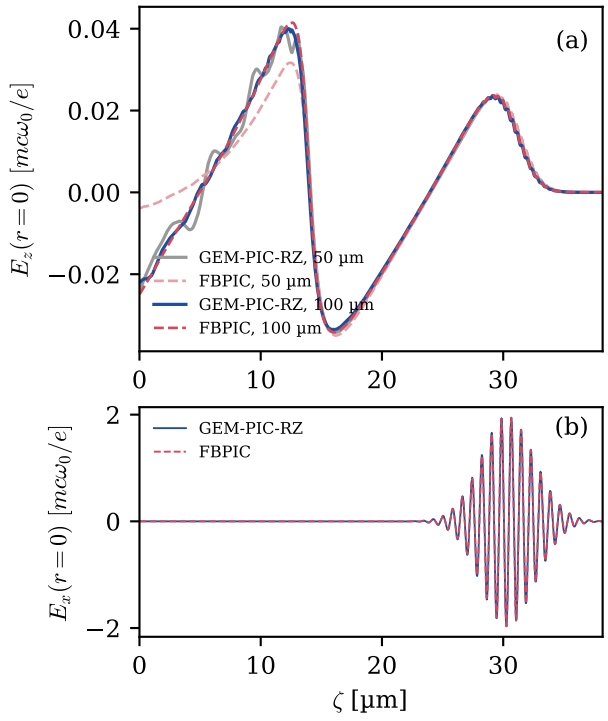


FIG. 1. Laser wakefield in the bubble regime ($a_0 = 2$, $0.002\,n_c$): (a) on-axis E_z after 50 and 100 μm of propagation, GEM-PIC-RZ (solid) and FBPIC (dashed); (b) the laser field E_x on the axis at 100 μm. The laser moves to the right.

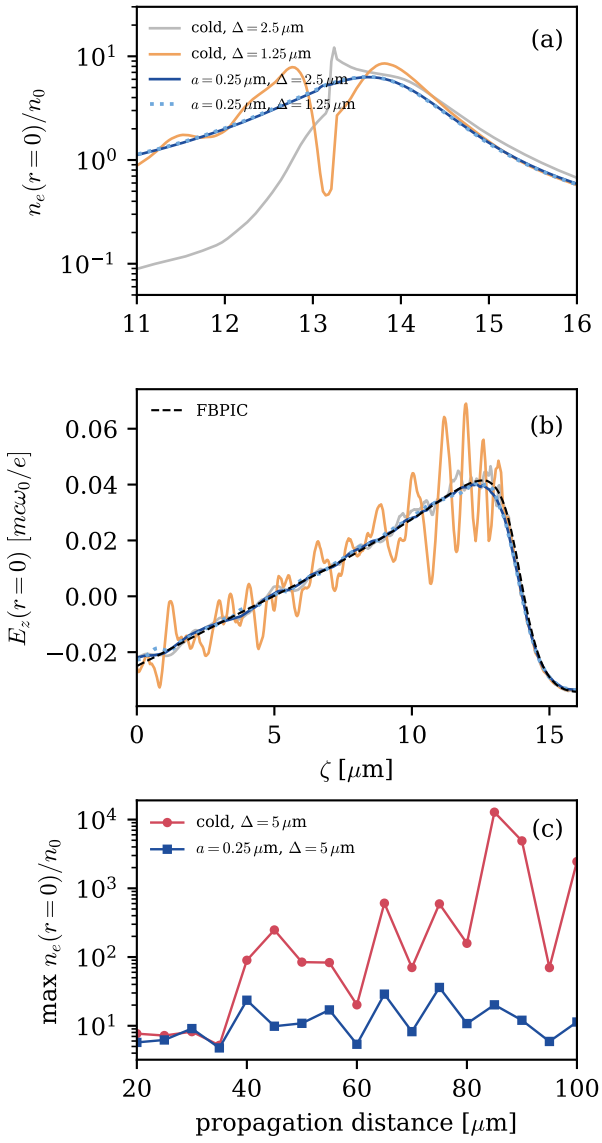


FIG. 2. Back of the first bubble in the case of Fig. 1 at 100 μm: (a) on-axis electron density, (b) on-axis E_z, cold plasma (grey, orange) and with source smoothing $a = 0.25\,\mu$m (blue), for $\Delta = 2.5$ and $1.25\,\mu$m, and FBPIC (dashed); (c) peak on-axis density at the closure against the propagation distance for $\Delta = 5\,\mu$m, cold (red) and smoothed (blue).

times the FBPIC time step $c\,\delta t = \lambda_0/32$), and the source smoothing $a = 2\,k_0^{-1}$ of Sec. III F.

Figure 1 shows the on-axis fields. After 100 μm the first wake period agrees to 1.9 % rms of the peak field, the minimum of E_z is -0.0335 vs $-0.0342\,m_e c\omega_0/e$, and the laser peak field 1.969 vs 1.954. In the up-ramp at 50 μm the codes differ by representation, not by physics: GEM-PIC-RZ shows the wake as it passes one plane, with the density of that plane, while FBPIC shows the wake behind the laser at smaller z, where the density is lower. The difference disappears where the density is constant over a window length. On one CPU core GEM-PIC-RZ took 1.7 min, FBPIC about 1.5 h.

B. Bubble closure

At the back of the bubble the expelled electrons return to the axis. There the accelerating field is strongest, there the electrons are injected, and there the witness bunch of a two-bunch scheme sits, so the fields in this region decide the quality of the accelerated beam. In a cold plasma with an axisymmetric driver the returning electron trajectories cross on the axis and form a caustic [9]: the density is singular, and on a fine radial grid it grows without bound as the cells are refined. Uniform grids of full PIC codes cut the singularity off implicitly over one cell; the fine axis cells of GEM-PIC-RZ expose it. The singular density moves the resonance of the reduced equations (Sec. III F) to resolved scales, and the step-dependent perturbations of the current excite it.

Figure 2 shows the result for the wake of Fig. 1. In the cold plasma the density at the closure is a narrow spike whose height and shape change with Δ [Fig. 2(a)], and the wake behind it carries oscillations on the scale of λ_0 that do not converge with Δ: the rms deviation from FBPIC is $1.8 \times 10^{-3}\,m_e c\omega_0/e$ for $\Delta = 2.5\,\mu$m and

9.8×10^{-3} for $\Delta = 1.25\,\mu$m [Fig. 2(b)]. With $\Delta = 5\,\mu$m the closure density alternates from step to step between about 70 and $10^4\,n_0$, the signature of the period-two modes, and the peak field reaches $0.86\,m_e c\omega_0/e$, twenty times the wake [Fig. 2(c)].

The spike is unphysical: real plasmas are warm, real drivers are not perfectly cylindrical, and the spike carries no energy. The smoothing of Eq. (25) with $a = 2\,k_0^{-1}$ ($0.25\,\mu$m, $0.09\,c/\omega_p$, about the axis cell) replaces it by a peak of finite width. The oscillations disappear: the rms deviation behind the bubble is $0.7 \times 10^{-3}\,m_e c\omega_0/e$ for both $\Delta = 2.5$ and $1.25\,\mu$m, and the two runs agree with each other to the same level, and with $\Delta = 5\,\mu$m the closure density stays bounded at 5–40 n_0. The price is a model change of order a at the bubble back: the peak of E_z there changes from 0.047 (the spike) to 0.040 (FBPIC: 0.042), and the rms agreement of the first wake period from 1.6 to 1.9 %. The criterion is that the fields behind the spike converge, not the height of the spike. In a denser plasma the closure is stronger and the wake needs a larger a (Sec. VI C). Two alternatives that also suppress the oscillations, an off-centred update of R in s and keeping electrons in their level for several steps, are available as options; the first damps the laser, the second changes the trapping (Sec. VI E).

C. Laser evolution: diffraction, depletion and red-shift

The energy gain of a laser wakefield stage is limited by the depletion of the laser, which transfers its energy to the wake and is red-shifted in the process [18, 19]. Following the laser over the full stage is the task for which GEM-PIC was designed, and it is where envelope models, which assume a slowly varying envelope, are most strained: for few-cycle pulses in dense plasmas, as used with kHz lasers, the pulse depletes within a few Rayleigh lengths and its spectrum changes by several percent. The configuration is such a case: $a_0 = 1.5$, $w_0 = 3.56\,\mu$m ($P/P_c = 0.56$), a field duration of 6.8 fs (2.5 laser cycles), starting at focus in a plasma of $0.01\,n_c = 1.74 \times 10^{19}\,\mathrm{cm}^{-3}$, followed over $300\,\mu$m. We compare with FBPIC and with the envelope code Wake-T [14], which combines a laser envelope solver with a quasi-static plasma model.

Over $300\,\mu$m the pulse diffracts, loses a quarter of its energy to the wake and is red-shifted by 4.5 % (Fig. 3). GEM-PIC-RZ ($h = 0.2\,k_0^{-1}$, $\Delta = 2\,\mu$m) agrees with FBPIC over the whole distance: the energy to $0.013\,W_0$, the red-shift to $0.002\,k_0$, the peak field to 5 % (13 % at the end, where the pulse is weak and spread out), and the ~~wake~~ first wake period on the axis to 6 % rms of its peak. Further behind, where the wake of the depleted pulse is weak, phase differences raise the rms deviation to about 20 %. With $\Delta = 1\,\mu$m the results change by less than $0.03\,W_0$. In the envelope model the pulse stays focused, drives a much stronger wake and loses only 5.5 % of its energy instead of 27 %; the Wake-T results do not change with resolution. GEM-PIC-RZ took 6.7 min on one core, FBPIC 3.7 h on two cores.

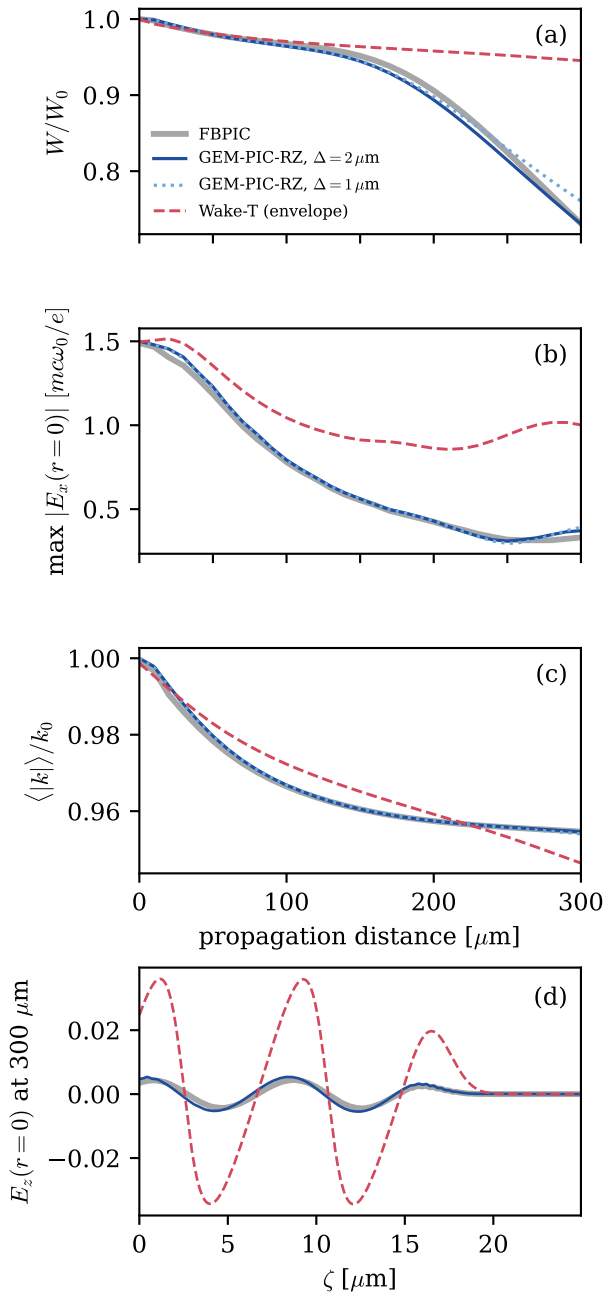


FIG. 3. Evolution of a 6.8 fs laser pulse ($a_0 = 1.5$) in a plasma of $0.01\,n_c$ over $300\,\mu$m (six Rayleigh lengths): (a) laser energy, (b) peak of $|E_x|$ on the axis, (c) spectral centroid of the laser fields; (d) on-axis E_z at $300\,\mu$m. FBPIC (grey), GEM-PIC-RZ with $\Delta = 2$ and $1\,\mu$m (blue) and the envelope code Wake-T (red, dashed).

At this density the closure of Sec. VI B is strong enough to make the cold-plasma run unstable several wake periods behind the laser, where E_z ~~reaches twenty times~~ exceeds the FBPIC value more than tenfold; the laser

itself is hardly affected ($W = 0.740\,W_0$). The wake ~~needs a~~ is stable for an $m = 0$ smoothing length of ~~about $0.3\,c/\omega_p$ ($3\,k_0^{-1}$)~~at least $2\,k_0^{-1}$ ($0.2\,c/\omega_p$); with $1\,k_0^{-1}$ it still deviates by 80 % rms at 300 μm, and lengths of 2, ~~while the~~ 3 and $4\,k_0^{-1}$ give the same laser and the same wake. The laser-driven $m = 1$ currents need ~~only $1\,k_0^{-1}$; smoothing them over $3\,k_0^{-1}$ would change~~ much less: with $a = 0.5$, 1 and $2\,k_0^{-1}$ the laser energy at 300 μm ~~by~~ is 0.726, 0.731 and 0.747 W_0, while $3\,k_0^{-1}$, a fifth of the laser wavelength, gives 0.773 W_0, 6 % above FBPIC. The run of Fig. 3 ~~therefore~~ uses $a = 1\,k_0^{-1}$ for $m \geq 1$ and $3\,k_0^{-1}$ for $m = 0$.

D. Ionization injection

Ionization injection [20, 21] is one of the most widely used ways to load electrons into a laser wakefield: a small admixture of a high-Z gas releases its inner-shell electrons near the peak of the laser field, inside the bubble, where they are trapped. The process depends on the instantaneous field of the carrier, which GEM-PIC-RZ resolves and an envelope code has to model with averaged rates, and it tests field ionization, trapping and acceleration together. The configuration is the case of Sec. VI A with $a_0 = 2.3$, still without self-injection, and N^{5+} ions at 1 % of the electron density, which both codes ionize to N^{6+} and N^{7+} by the ADK rate.

The ionized fraction of the dopant agrees (3.1 % N^{6+} and 0.3 % N^{7+} behind the laser), and so does the injected bunch (Fig. 4). At $z = 183\,\mu$m its charge above 5 MeV is 1.58 pC against 1.76 pC in FBPIC, its mean energy 12.4 against 12.6 MeV and its rms energy spread 5.2 against 5.4 MeV, and the spectra agree up to the highest energy. Halving Δ changes the charge by 7 %. Below about 4 MeV FBPIC holds more electrons: sheath electrons that are not trapped and that GEM-PIC-RZ streams through the window. ~~Without the source smoothing GEM-PIC-RZ traps 30 % more charge, a measure of how strongly the closure affects the injection~~The injected charge, unlike its energy, depends on the smoothing length, because the trapping is decided at the closure where the smoothing acts (Sec. VI F). GEM-PIC-RZ took 8.5 min on one core, FBPIC 3.4 h on two.

E. Density down-ramp injection: an open problem

In a density down-ramp the plasma wavelength grows, the back of the bubble slows down, and background electrons that it overtakes are trapped [22–24]. Sharp ramps, made with shocks or blades in a gas jet, give bunches with small energy spread and are used in many experiments. We report this benchmark as a caution, because GEM-PIC-RZ does not yet pass it. The laser of Sec. VI A, now with $a_0 = 1.8$, crosses a plateau of $2n_0$ ($n_0 = 0.002\,n_c$) that drops linearly to n_0 between $z = 60$ and 70 μm. Behind the ramp the bubble lengthens by about 30 % within 10 μm, and its back moves forward through the window at about $0.5\,c$.

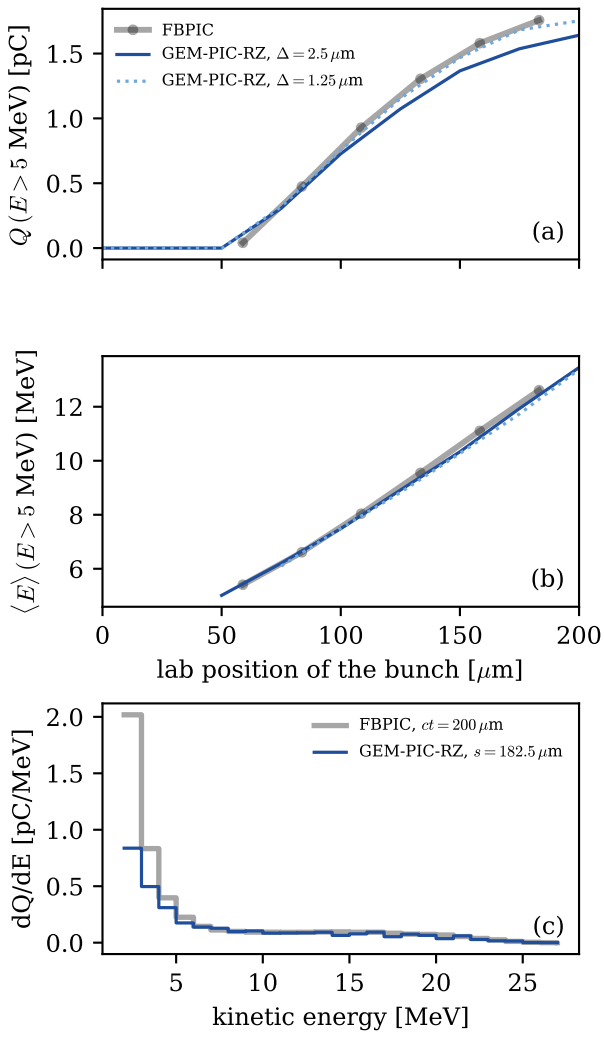


FIG. 4. Ionization injection from a nitrogen dopant ($a_0 = 2.3$, $0.002\,n_c$, 1 % N^{5+}): (a) charge and (b) mean energy of the injected electrons above 5 MeV against the lab position of the bunch, (c) their spectrum at $z = 183\,\mu$m. FBPIC (grey) and GEM-PIC-RZ with $\Delta = 2.5$ and 1.25 μm (blue).

Qualitatively the codes agree: both start trapping at the ramp (on a constant plateau of $2n_0$ GEM-PIC-RZ traps no electrons), and the wake on the plateau before the ramp agrees to $2 \times 10^{-3}\,m_e c\omega_0/e$ rms. Quantitatively they do not (Table III). FBPIC traps 51 pC above 2 MeV; GEM-PIC-RZ traps between 9 and 28 pC, and the charge depends on Δ without converging. The difference persists with a finer step inside the ramp alone, with a higher velocity threshold for the beam list and with stronger source smoothing; keeping newly trapped electrons on the streaming list for several steps suppresses the trapping almost entirely.

TABLE III. Density down-ramp injection: charge (pC) of the trapped electrons above 2 and 5 MeV. FBPIC at $ct = 150$ and $200\,\mu$m (bunch at $z = 136$ and $179\,\mu$m); GEM-PIC-RZ at the levels $s = 150$ and $200\,\mu$m. The last row uses $\Delta = 0.125\,\mu$m in the ramp and $1.25\,\mu$m elsewhere.

	ct or $s = 150\,\mu$m		ct or $s = 200\,\mu$m	
	> 2 MeV	> 5 MeV	> 2 MeV	> 5 MeV
FBPIC	51.0	14.6	51.1	30.7
$\Delta = 2.5\,\mu$m	9.9	8.9	8.5	8.4
$\Delta = 1.25\,\mu$m	20.9	17.1	19.6	18.8
$\Delta = 0.5\,\mu$m	12.2	9.9	–	–
$\Delta = 0.25\,\mu$m	14.2	13.7	–	–
ramp refined	–	–	27.6	12.6

We attribute the discrepancy to the choice of GT2 for the particles. In GT2 a level is a plane $z = s$, and an electron is trapped when its lab position reaches the next plane inside the window; the decision is made in the fields of one plane, while the back of the bubble, which does the trapping, moves through the window by a large fraction of its length between two planes. In GT1 a level is an instant of time, as in FBPIC, and the moving back of the bubble is followed in time; this frame should describe down-ramp injection. GT1 has its own difficulty, however: plasma particles that reach the next slow-time level inside the window stop being pushed there, and this imprints stripes on the fields. A particle scheme that combines the two will be the subject of a follow-up study. Until then, GEM-PIC-RZ results for self-injection in sharp density transitions should not be trusted quantitatively, whereas ionization injection (Sec. VI D), where the electrons are born inside a slowly evolving bubble, is reproduced.

F. Robustness to the smoothing length and the step

The source smoothing is a numerical model parameter, and a result is only as good as its independence of it. We therefore repeated every benchmark with smoothing lengths from 0 to $4\,k_0^{-1}$ and with half the step (Table IV; scripts and runs in the repository). The conclusions fall into three classes.

Robust. The wake up to the back of the first bubble agrees with FBPIC to 1.6–2.1 % for all $a \le 2\,k_0^{-1}$ and both steps. The laser energy and red-shift after six Rayleigh lengths agree to within $0.016\,W_0$ and $0.0012\,k_0$ for all $m \ge 1$ lengths up to $2\,k_0^{-1}$ and all $m = 0$ lengths that keep the wake stable, and the half step changes the energy by $0.03\,W_0$; even the unstable cold run gives the laser correctly. The energy of the ionization-injected electrons (11.7–12.3 MeV against 12.1 MeV) does not depend on $a \le 2\,k_0^{-1}$ or Δ. These conclusions do not rest on the choice of a.

Bounded by the choice of a. The wake behind the closure needs a minimum smoothing length. Below it the result does not converge with Δ (cold: 1.8 against 9.7×10^{-3}; $a = 1\,k_0^{-1}$: 1.1 against 1.8×10^{-3}); at $a = 2\,k_0^{-1}$ it converges and agrees best with FBPIC, at $4\,k_0^{-1}$ it converges with a larger model error, and the field at the bubble back drops to 0.036 (FBPIC 0.042). In the denser plasma of Sec. VI C the minimum is $2\,k_0^{-1}$ for $m = 0$, about $0.2\,c/\omega_p$ against $0.09\,c/\omega_p$ at $0.002\,n_c$, and the laser-driven currents tolerate up to $2\,k_0^{-1}$. The lengths used in the figures are the smallest that converge, and the results quoted for them hold within the stated agreement for a factor of two around them.

Sensitive. The injected charge depends on a because trapping is a threshold process decided at the closure, where the smoothing acts. Above 5 MeV it ranges from 1.99 to 1.54 pC for $a = 0$ to $2\,k_0^{-1}$, with FBPIC (1.70 pC) inside this range for both steps, and drops to 0.96 pC at $a = 4\,k_0^{-1}$. The agreement of Sec. VI D in charge is therefore to be read with a model uncertainty of about ± 15 %, larger than the 7 % from the step; the energy and the spectrum are robust. For the down-ramp the table shows the converse: for every a and Δ GEM-PIC-RZ traps 10–22 pC against 51 pC, so the discrepancy of Sec. VI E is not caused by the regularization.

VII. UNDULATOR RADIATION AND FREE-ELECTRON LASERS

Because the carrier is resolved, the same equations describe the radiation of the accelerated beam. A beam that wiggles in an undulator radiates through the transverse current in Eqs. (3), (4), and the radiation acts back on the beam; it stays at fixed ζ while the electrons slip back by λ_r per period. Spontaneous and coherent emission and the FEL interaction are therefore contained in the model, with the full space-charge fields of the beam and without the period averaging of dedicated FEL codes [37]. The ζ grid resolves λ_r, the step in s the undulator period. First tests are encouraging: the coherent spontaneous emission of a short bunch agrees with the far-field radiation integral to 2–4 % in energy, the SASE gain length of a compact FEL with the Ming Xie fit [38] (6.3 vs 6.4 periods), and seeded amplifiers at 10 nm and 0.8 nm with GENESIS 1.3 [37] to 4 % and 0.6 % in gain length. A combined simulation of plasma acceleration, beam transport and FEL amplification in one code will be presented in a separate publication.

VIII. CONCLUSIONS

GEM-PIC-RZ brings the Galilean electromagnetic PIC scheme to quasi-cylindrical geometry. It resolves the laser carrier, as a full PIC code, and has no Courant limit, as

TABLE IV. Sensitivity of the benchmark results to the source-smoothing length a (in k_0^{-1}) and the step Δ. Pairs give the result for the step of the benchmark and for half of it ($\Delta = 2.5/1.25\,\mu$m; laser: $2/1\,\mu$m). Laser evolution: a is the length for $m \geq 1$, with $3\,k_0^{-1}$ for $m = 0$ (cold: both zero); with $a = 3$ for $m \geq 1$, $W/W_0 = 0.773$. Ionization injection: bunch at $z = 175\,\mu$m, FBPIC interpolated to it. Fields in $m_e c\omega_0/e$, deviations from FBPIC rms, in % of the peak or in $10^{-3}\,m_e c\omega_0/e$. Bold: the settings of the benchmark figures.

benchmark	quantity	FBPIC	$a = 0$	$a = 1$	$a = 2$	$a = 4$
wake, 100 μm	rms dev., first period [%]	–	1.6 / 2.0	1.8 / 2.0	**1.9** / 2.1	3.4 / 3.6
	rms dev., behind bubble [10^{-3}]	–	1.8 / 9.7	1.1 / 1.8	**0.70** / 0.72	1.5 / 1.9
	E_z at bubble back	0.042	0.047 / 0.069	0.042 / 0.042	**0.040** / 0.041	0.036 / 0.035
laser, 300 μm	W/W_0	0.731	0.740	**0.731** / 0.761	0.747	–
	$\langle\|k\|\rangle/k_0$	0.9547	0.9549	**0.9546** / 0.9541	0.9535	–
	rms dev., wake [%]	–	1300	**19** / 24	16	–
ionization inj.	$Q(E > 5\,\mathrm{MeV})$ [pC]	1.70	1.99	1.84 / 1.95	**1.54** / 1.68	0.96
	$\langle E\rangle$ [MeV]	12.1	12.3	12.1 / 12.0	**11.9** / 11.7	11.0
down-ramp inj.	$Q(E > 2\,\mathrm{MeV})$, 150 μm [pC]	51.0	18.8 / 21.9	14.4	**9.9** / 20.9	10.1 / 17.0

a quasi-static code, so its step is set by the evolution of the laser and the plasma, not by the cell size; in the laser wakefield benchmarks it is a hundred times the time step of the full PIC code. Two numerical problems of the scheme were found and solved: a stationary odd–even mode that radiates behind relativistic bunches, removed by a filter of the beam current, and the excitation of spurious modes by the singular bubble closure of a cold plasma, removed by a charge-conserving radial smoothing of the plasma sources.

Against the full PIC code FBPIC, GEM-PIC-RZ reproduces the bubble-regime wake, the depletion and redshift of a few-cycle pulse over six Rayleigh lengths, where an envelope model fails, and ionization injection, in charge, energy and spectrum, at about 2 % of the CPU time. This makes it suited for parameter scans and for the design of laser wakefield stages over their full length. These conclusions hold for smoothing lengths within a factor of two around the values used and for half the step; the injected charge is the exception, with a model uncertainty of about 15 % from the regularization. Two limits remain. Backward-propagating radiation (Raman backscatter, reflection from overdense plasma) is outside the model, since $\partial_s L$ is dropped. Self-injection in a sharp density down-ramp is not yet reproduced; we attribute this to the GT2 frame of the particles and will address it, together with the artifacts of the GT1 frame, in a follow-up study. Planned work also includes benchmarks of beam-driven wakes, GPU performance measurements and the FEL applications.

GEM-PIC-RZ is open source [40], with the decks and scripts of all tests in this paper. Its development, like that of QUARZ, shows that a production-grade simulation code can be built by an AI agent under the direction of a physicist, when every feature is held to an independent test.

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